

EMMANUEL VARGAS FLORES

SYSTEMS INTEGRATION ENGINEER | INDUSTRIAL AUTOMATION | SOFTWARE & IT SYSTEMS

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PROFESSIONAL SUMMARY

Systems Integration Engineer and Systems Automation Lead with experience across industrial automation, manufacturing systems, enterprise platforms, software, APIs, databases, sensors and IT infrastructure. Skilled at translating operational requirements into system designs, coordinating multidisciplinary stakeholders, managing configuration and change impacts, and guiding solutions through implementation, testing, validation, deployment and operational handover. Background in regulated medical-device manufacturing. Fluent in English and Spanish; open to international travel and relocation to the Netherlands.

CORE COMPETENCIES

Systems Integration: Architecture, interfaces, requirements traceability, configuration and change impact.	Testing & Delivery: Functional testing, validation, troubleshooting, deployment and monitoring.
Software & APIs: C#/.NET, Node.js/Express, Python, JavaScript, React, REST APIs and microservices.	Industrial Systems: Manufacturing applications, IoT sensors, production monitoring, automation and RPA.
Engineering Practices: Risk identification, technical documentation, software validation and Agile/Scrum.	Stakeholder Leadership: Cross-functional coordination with engineering, operations, quality and business teams.

PROFESSIONAL EXPERIENCE

Systems Integration Engineer / Systems Automation Lead | AVNA

Grecia, Costa Rica | June 2025 - Present

- Lead integration and automation work from requirements definition through implementation, validation, deployment and operational support.
- Develop enterprise integrations, internal applications and microservices using APIs and databases.
- Build and maintain software solutions, with attention to authentication, logging, documentation and deployment practices.
- Coordinate stakeholders, assess risks and change impacts, troubleshoot integration issues and support operational handover.
- Prepare technical specifications, work instructions and software validation documentation in a regulated manufacturing environment.

IT Support Technician Intern / Software Development | AVNA

Grecia, Costa Rica | February 2025 - April 2025

- Developed a business-process application using React, REST services and Node.js.
- Implemented routing, input validation, responsive interfaces and functional/usability testing while supporting users and production IT needs.

Automation Technician | MicroVention (Terumo Neuro)

Alajuela, Costa Rica | August 2019 - July 2022

- Developed, validated and maintained C#/.NET software for medical-device manufacturing, supporting reliability and controlled change.
- Built UiPath RPA solutions to improve process consistency and reduce repetitive work.
- Supported production systems, labeling, automation equipment and material-movement initiatives in collaboration with Engineering and Operations.
- Troubleshoot software and automation issues, documented solutions and translated operational requirements into technical improvements.

SOLUTIONS I CAN BUILD

- **Systems integration:** Connect business applications through APIs and microservices.
- **Manufacturing dashboards:** Build monitoring and reporting applications for operational visibility.
- **Process automation:** Automate repetitive workflows and data collection.
- **AI assistants:** Integrate language models into business applications.
- **Tracking systems:** Develop solutions for purchase orders, materials and operational workflows.
- **Industrial IoT:** Connect sensors with monitoring and reporting applications.

EDUCATION

Bachelor of Science in Informatics Engineering

Universidad Estatal a Distancia (UNED), Costa Rica
Completed 2026 | SINAES-accredited program

Diploma in Informatics

Universidad Estatal a Distancia (UNED) | 2019 - 2023

Computer and Informatics Technician

Universidad Estatal a Distancia (UNED) | 2018 - 2019

TECHNICAL SKILLS

Programming: C#, .NET/ASP.NET Core, Node.js, Express, Python, JavaScript, C++, PHP

Frontend: React, Vite, JSX, HTML, CSS

Integration & Data: REST APIs, JSON, Plex Connect, Microsoft Graph, Dataverse, SQL Server, MySQL, MariaDB, PostgreSQL

Infrastructure: Linux/Ubuntu Server, Docker, Nginx, PM2, Git, on-premise deployments, CI/CD concepts

Automation: Industrial automation, IoT/sensors, UiPath RPA, scheduled jobs, production monitoring, computer-vision exposure

LANGUAGES & MOBILITY

Spanish: Native | English: C1 - Advanced professional proficiency

Available for relocation to the Netherlands and frequent international travel.